

Hairuo Fu

LunaSCOPE Postdoctoral Fellow

Dept. of Earth, Environmental and Planetary Sciences,

Brown University, Providence, RI 02912

+1 857-253-1968

Email: hairuo_fu@brown.edu

Homepage: fuhairuo.github.io

Academic positions

- 2026 Aug.– Assistant Project Scientist, UCLA
Independent research; managing MC-ICP-MS and clean labs
(mentor: Peng Ni)
- 2024–2026 LunaSCOPE Postdoctoral Fellow, Brown University
(mentors: James Dottin & Alex Evans)
- 2020–2024 Teaching Fellow, Harvard University
- 2019–2024 Research Assistant, Harvard University

Education

- 2018–2024 Ph.D., Isotope Geochemistry, Harvard University
Dissertation: Uncovering Geochemical Evidence for the Moon's High-energy Giant-impact Origin
(advisor: Stein B. Jacobsen)
- 2018 Visiting student at NERC Isotope Geosciences Laboratories, British Geological Survey, UK
(mentor: Daniel J. Condon)
- 2016–18 Graduate research, Geology, China University of Geosciences, Beijing (CUGB)
(advisor: Shihong Zhang)
- 2016 BSc Honors, Geology, China University of Geosciences, Beijing (CUGB)
Dissertation: New paleomagnetic results from the Neoproterozoic mafic sills in the Sibao area
(advisor: Shihong Zhang)

Fellowships & Awards

- Research fund awarded by LunaSCOPE (\$10,000, 2024–2026)
- LunaSCOPE Postdoctoral Fellowship, Brown University (\$130,000 for stipend, 2024–2026)
- Fellow of Origins of Life Initiative (2020–2024)
- Department Fellowship, Harvard University (2018–2019)
- CUGB Innovation Fund for overseas research (\$15,000, 2017–2018)
- CUGB Professional Scholarship (2012–2016)
- Fellow of the 1st Innovation Class of CUGB (2013–2016)
- CUGB LongRun Scholarship (2013)

Manuscripts in-revision & under-review

1. **Fu, H.**, Dottin III, J.W., Sulfur geochemical evidence for a high-energy impact lunar origin (*under review*) (<https://www.researchsquare.com/article/rs-6762675/v1>).
2. **Fu, H.**, Jacobsen, S. B., Larsen, B. T. The ⁴⁰K-⁴⁰Ca chronometer as a tracer of magma sources and crustal contamination (*under review*).

Manuscripts in-preparation

1. **Fu, H.**, Parmentier, E. M., Huber, C., Melt Migration in Early Lunar Mantle and Its Influences on Lunar Chemical Differentiation (prospectively 2026).
2. **Fu, H.**, Advancing Giant-Impact Hypothesis Testing: Integrating Impact Simulations with Pre-Impact Body Differentiation (prospectively 2026).

Publications

1. Patil, K., Dottin III J. W., **Fu, H.**, Monteleon, B., Chatterjee, N., Hooper, N., Izon, G., Adams J. C., Irving, A., Shearer, C. K. Franz, H. B., S Ono, S., 2026 (in press), Isotopic evidence for ancient exotic sulfur delivery to the Martian mantle, *Communications Earth & Environment*.
2. **Fu, H.**, Jacobsen S. B., 2025, Lunar refractory element evidence challenges the canonical giant-impact hypothesis. *Earth and Planetary Science Letters* 672, 119697. (<https://doi.org/10.1016/j.epsl.2025.119697>)
3. Dottin III, J.W., Simon S. B., Shearer C. K., Benson J., **Fu, H.**, Boesenberg J. S., Monteleone B., and the ANGSA Science Team, 2025, Endogenous, yet exotic, sulfur in the lunar mantle. *Journal of Geophysical Research: Planets* 130, e2024JE008834 (<https://doi.org/10.1029/2024JE008834>)
4. **Fu, H.**, Jacobsen S. B., 2024, Earth-Moon refractory element similarity constrains a thoroughly-mixed Moon-forming disk. *Earth and Planetary Science Letters* 646, 119008. (<https://doi.org/10.1016/j.epsl.2024.119008>)
5. Loeb, A., Jacobsen, S. B., Tagle, R., Adamson, T., Bergstrom S., Cloete, R., Cohen, S., Conrad, K., Domine, L., **Fu, H.** et al., 2024, Discovery of Spherules of Likely Extrasolar Composition in the Pacific Ocean Site of the CNEOS 2014-01-08 (IM1) Bolide. *Chemical Geology* (<https://doi.org/10.1016/j.chemgeo.2024.122415>)
6. **Fu, H.**, Jacobsen, S. B., Sedaghatpour, F., 2023, Moon's high-energy giant-impact origin and differentiation timeline inferred from Ca and Mg stable isotopes. *Communications Earth & Environment*. 4, 307 (<https://doi.org/10.1038/s43247-023-00974-4>)
7. **Fu, H.**, Zhang S., Condon, D. J., Xian H., 2022, Secular change of true polar wander over the past billion years. *Science Advances* 8, eabo2753. (science.org/doi/10.1126/sciadv.abo2753)
8. **Fu, H.**, Jacobsen, S. B., Larsen, B. T., Eriksen, Z. T., 2022, Ca-isotopes as a robust tracer of magmatic differentiation. *Earth and Planetary Science Letters* 594, 117743. (doi.org/10.1016/j.epsl.2022.117743)
9. Park, Y., Swanson-Hysell, N. L., Xian, H., Zhang, S., Condon, D. J., **Fu, H.**, Macdonald, F. A., 2021, A consistently high-latitude South China from 820 to 780 Ma: Implications for exclusion from Rodinia and the feasibility of large-scale true polar wander. *Journal of Geophysical Research: Solid Earth*, 126, e2020JB021541. (doi.org/10.1029/2020JB021541)
10. Ren, Q., Zhang, S., Wu, Y., Yang, T., Gao, Y., Turbold, S., Zhao, H., Wu, H., Li H., **Fu, H.**, Xu, B., Zhang, J., Tomurtogoo, O., 2018, New late Jurassic to early Cretaceous paleomagnetic results from North China and southern Mongolia and their implications for the evolution of the Mongol-Okhotsk suture. *Journal of Geophysical Research: Solid Earth*, 123. (<https://doi.org/10.1029/2018JB016703>)

Conference abstracts

1. **Fu, H.**, Parmentier, E. M., Huber, C., Melt Migration in Early Lunar Mantle and Its Influences on Lunar Chemical Differentiation. 57th Lunar and Planetary Science Conference (#1192) (<https://www.hou.usra.edu/meetings/lpsc2026/pdf/1192.pdf>)
2. **Fu, H.**, Boukaré, C.-E., Nakajima, M., Advancing Giant-Impact Hypothesis Testing: Integrating Impact Simulations with Pre-Impact Body Differentiation. 57th Lunar and Planetary Science Conference (#1368) (<https://www.hou.usra.edu/meetings/lpsc2026/pdf/1368.pdf>)

3. **Fu, H.**, Parmentier, E. M., Huber, C., Melt Retention and Migration in Lunar Cumulates Reconcile KREEP Composition and Crustal Chemical Heterogeneity. Abstract for AGU 2025 fall meeting (<https://agu.confex.com/agu/agu25/meetingapp.cgi/Paper/1874052>)
4. **Fu, H.**, Boukaré, C.-E., Nakajima, M., Advancing Giant-Impact Hypothesis Testing: Integrating Impact Simulations with Pre-Impact Body Differentiation. Abstract for AGU 2025 fall meeting (<https://agu.confex.com/agu/agu25/meetingapp.cgi/Paper/1936393>)
5. **Fu, H.**, Dottin III, J.W., 2025, [Geochemical constraints on the origin of the Moon: Insights from sulfur concentration and isotope coevolution models](#). NASA Exploration Science Forum
6. **Fu, H.**, Dottin III, J.W., 2025, Post-Giant-Impact Lunar Sulfur and Sulfur Isotope Coevolution: Insights into the Moon's Origin. 55th Lunar and Planetary Science Conference (LPI Contrib. No. 1616). (<https://www.hou.usra.edu/meetings/lpsc2025/pdf/1616.pdf>)
7. **Fu, H.**, Jacobsen, S. B., Larsen, B. T., 2024, The ⁴⁰K-⁴⁰Ca chronometer as a tracer of magma sources and crustal contamination. Goldschmidt 2024 (<https://conf.goldschmidt.info/goldschmidt/2024/meetingapp.cgi/Paper/24343>)
8. **Fu, H.**, Jacobsen, S. B., 2024, Moon's Predicted Chemical Composition under Canonical Giant-impact Models and Its Compatibility with Lunar Observations. 55th Lunar and Planetary Science Conference (LPI Contrib. No. 2054). (<http://www.hou.usra.edu/meetings/lpsc2024/pdf/2054.pdf>)
9. Loeb, A., Jacobsen, S. B., Adamson, T., Bergstrom, S., Cloete, R., Cohen, S., Conrad, K., Domine L., **Fu, H.**, et al., 2024, Spherules Recovered from the Pacific Ocean Site of the CNEOS 2014-01-08 (IM1) Bolide. 55th Lunar and Planetary Science Conference (LPI Contrib. No. 2130). (<https://www.hou.usra.edu/meetings/lpsc2024/pdf/2130.pdf>)
10. Fu, R.R., Thaler, P., Maxemin, G., Steele S.C., **Fu, H.**, Heck, P.R., 2024, A Strongly Magnetic Sub-Region of the Solar Nebula Defined by the CM, CO, and CV Chondrites. 55th Lunar and Planetary Science Conference (LPI Contrib. No. 1511) (<https://www.hou.usra.edu/meetings/lpsc2024/pdf/1511.pdf>)
11. Jacobsen, S. B., Parendo, C. A., Eriksen, Z. T., **Fu, H.**, Gérard, Y., 2023, The Nu Sapphire SP001 collision cell MC-ICP mass spectrometer: Application to high-precision measurements of K and Ca isotopes. Goldschmidt 2023 (conf.goldschmidt.info/goldschmidt/2023/meetingapp.cgi/Paper/20550)
12. Fu, R. R., Thaler P, Maxemin G., Steele S., **Fu, H.**, Solar nebula magnetic fields recorded in chondrules from the Murchison CM chondrite. Abstract for AGU 2023 fall meeting (<https://agu.confex.com/agu/fm23/meetingapp.cgi/Paper/1392285>)
13. **Fu, H.**, Jacobsen, S. B., Sedaghatpour, F., 2023, Timeline of lunar differentiation recorded in Ca and Mg stable isotopes. 54th Lunar and Planetary Science Conference (LPI Contrib. No. 2806). (<http://www.hou.usra.edu/meetings/lpsc2023/pdf/2400.pdf>)
14. **Fu, H.**, Jacobsen, S. B., 2023, Testing the Earth-Moon Synestia with trace elements. 54th Lunar and Planetary Science Conference (LPI Contrib. No. 2806). (<https://www.hou.usra.edu/meetings/lpsc2023/pdf/2683.pdf>)
15. Zhao, H., Zhang, S., **Fu, H.**, Yang, K., Li, H., Wu, H. and Yang, T., 2023, Characterizing the geomagnetic paleosecular variation at high latitudes in late Mesoproterozoic: new evidence from carbonate successions of the Jingeryu Formation in the North China craton. Abstract for AGU 2022 fall meeting (<https://agu.confex.com/agu/fm22/meetingapp.cgi/Paper/1084265>)
16. Chang, L., Zhang, S., Xian, H., **Fu, H.**, and Li, H., 2021, Paleomagnetic insights into the Neoproterozoic connection between South China and India, and their position in Rodinia. Abstract for AGU 2021 fall meeting (<https://agu.confex.com/agu/fm21/meetingapp.cgi/Paper/994801>)
17. **Fu, H.**, Jacobsen, S. B., Larsen, B. T., Eriksen, Z. T., 2022, Ca-isotopes as a robust tracer of magmatic differentiation. Goldschmidt 2022

(<https://conf.goldschmidt.info/goldschmidt/2022/meetingapp.cgi/Paper/12513>)

18. Ren, Q., Zhang, S., Zhao, H., Ding, J., Turbold, S., Gao, Y., Xu, B., Wu, Y., **Fu, H.**, 2017, New late Jurassic to early Cretaceous paleomagnetic results from North China and southern Mongolia and their implications for the evolution of the Mongol-Okhotsk suture. Abstract for AGU 2017 fall meeting (<https://agu.confex.com/agu/fm17/meetingapp.cgi/Paper/220039>)
19. Xian, H., Zhang, S., Xiao, Q., Li, H., Chang, L., **Fu, H.**, Liu, R., 2017, Tonian paleomagnetism of the red beds of Madiyi Formation, lower Banxi Group in South China: implications for pre-Sturtian climate, Rodinia reconstruction and true polar wander. Abstract for AGU 2017 fall meeting.

Teaching

- ESE6–Introduction to Environmental Science and Engineering, Harvard University (TF, undergrad level, Profs. Steven Wofsy and Elsie Sunderland, 2020 Spring)
- GENED 1018: How to Build a Habitable Planet, Harvard University (TF, general education for undergrad, Prof. Charles Langmuir, 2020 Fall)
- EPS240–Isotope and Trace Element Geochemistry and Geochronology, Harvard University (TF, grad level, Prof. Stein Jacobsen, 2022 Spring)
- EPS141–Geochemical and Cosmochemical Thermodynamics, Harvard University (TF, grad level, Prof. Stein Jacobsen, 2023 Spring)
- EPS110–Introduction to Planetary Materials and Earth Resources (TF, undergrad level, Prof. Stein Jacobsen, 2023 Fall)

Talks

- **[Invited]** Stable and radiogenic Ca isotopes in magmatic systems: Rb–Sr meets a challenger (UCLA, Jan. 29, 2026)
- Melt migration in early lunar mantle and its influences on lunar chemical differentiation (Geochemistry, Magma, and Petrology research seminar, Brown University, Oct. 31, 2025)
- Paleomagnetic perspectives on the coupling of true polar wander and mantle thermal evolution (Geophysics research seminar, Brown University, Sep. 30, 2025)
- Sulfur geochemical evidence for a high-energy impact lunar origin (LunaSCOPE Annual Meeting, Brown University, Aug. 6, 2025)
- **[Invited]** From magma chambers to magma oceans: novel geochemical insights into planetary differentiation and Earth-Moon formation (Colloquium, The University of Hong Kong, Apr. 7, 2025)
- **[Invited]** “Non-traditional” stable isotopes and their applications in high-temperature geochemistry (guest lecture at Environmental Stable Isotopes, Brown University, Apr. 1, 2025)
- Lunar origin and chronology inferred from Ca, Mg, and S isotopes (Geochemistry, Magma & Petrology research seminar, Brown University, Mar. 8, 2025)
- Moon’s predicted chemical composition under canonical giant-impact models and its compatibility with lunar observations (Brown Postdoctoral Research Symposium, Mar. 27, 2025)
- A new approach to determine lunar refractory element compositions: implications for Moon formation (Geochemistry, Magma & Petrology research seminar, Brown University, Nov. 15, 2024)
- **[Invited]** Geochemical evidence for the giant-impact formation of the Earth-Moon system (guest lecture at Evolution of the Moon seminar, Brown University, Sep. 11, 2024)
- **[Invited]** Testing Moon-forming giant-impact models with lunar chemical and isotopic compositions (LunaSCOPE Annual Meeting, Brown University, Aug. 6, 2024)
- Making the Earth-Moon system with a canonical or high-energy giant impact? A geochemical

perspective (Graduate Student and PostDoc seminar, Harvard University, Oct. 31, 2023)

- Planetary reorientation (true polar wander) and new interpretations of its evolution on Earth (Graduate Student and PostDoc seminar, Harvard University, Nov. 1, 2022)
- Probing Solar nebula magnetic fields: paleomagnetic studies on the Murchison meteorite (First-year symposium, Harvard University, Sep. 3, 2019)

Academic & professional service

- LPSC 2026 session chair ([From Theia to the Magma Ocean: Isotopic and Geophysical Constraints on Early Lunar Evolution](#))
- AGU 2025 session convener ([Planetary Impacts as a Fundamental Geophysical Process: Multidisciplinary Perspectives](#))
- AGU 2025 session chair ([Planetary Impacts as a Fundamental Geophysical Process: Multidisciplinary Perspectives](#))
- AGU 2025 session chair ([To the Moon: A New Era of Lunar Science III: Geology and Geophysics V Oral](#))
- External reviewer for the NASA Emerging Worlds proposal
- Panelist on *How to Land a Postdoc in STEM* (Harvard University, Sep. 18, 2024)
- Peer reviewer for:
 - Geochimica et Cosmochimica Acta
 - Geophysical Research Letters
 - Earth and Planetary Science Letters
 - Chemical Geology
 - Journal of Geophysical Research: Solid Earth
 - Geochemistry, Geophysics, Geosystems
 - Lithos
 - Space: Science & Technology
 - SCIENCE CHINA Earth Sciences

Field Trips

- Introduction to Planetary Materials and Earth Resources course field trip, Boston area (10/2023)
- Field sampling for igneous and metamorphic complex in the Oslo Rift, Norway (one week, 10/2022)
- Introduction to Planetary Materials and Earth Resources course field trip, Oslo, Norway (one week, 10/2022)
- Paleomagnetic sampling for Archean volcanic rocks, Pilbara, West Australia (three weeks, 07/2018)
- Paleomagnetic sampling for Neoproterozoic layered intrusions (ca. 830 Ma), Fanjingshan, Guizhou, South China (Field lead, four weeks, 07/2016)
- Paleomagnetic sampling for Neoproterozoic layered intrusions, Sibao, Guangxi, South China (Field lead, four weeks, 10/2015)
- Paleomagnetic sampling for Neoproterozoic sedimentary rocks, Zigui, Hubei, South China (three weeks, 06/2015)

Outreach

- [New Brown study may upend theories on moon's formation](#) (interview with The Brown Daily Herald)
- [International Observe the Moon Night: Museum of Natural History and Planetarium x LunaSCOPE](#)

- (scientist participant, Roger Williams Park, Providence, Oct. 4 to 5, 2025)
- International Observe the Moon Night: Museum of Natural History and Planetarium x LunaSCOPE (scientist participant, Roger Williams Park, Providence, Spt. 14, 2024)
 - Introducing Earth and planetary sciences to high-school students in remote areas of China (online seminar series, Jun. 26, 2024)
 - [Did 'alien' debris hit Earth? Startling claim sparks row at scientific meeting](#) (talk featured by *Nature Magazine*, Mar. 13, 2024)
 - [Planet+AI program](#) (member of AI Experiences with Planetary Data Summer Course, 2024)
 - Invited talk for high-school students: Lunar Narrative Through Apollo Petrographic Thin Sections (Planet+AI Webinar Series, Jan. 26, 2024)
 - Judge of the 2024 National Collegiate Research Conference (Jan. 20–21, 2024)